

SHETH LUJ & MV COLLEGE

PRACTICAL NO 4

AIM : Digital Signature

Implement digital signature algorithms such as RSA-based signatures, and verify the integrity and authenticity of digitally signed messages.

NORMAL CODE :

```
from cryptography.hazmat.primitives import hashes
from cryptography.hazmat.primitives.asymmetric import padding, rsa
from cryptography.exceptions import InvalidSignature
```

```
def create_rsa_keys():
    """Generate an RSA private key and corresponding public key."""
    private_key = rsa.generate_private_key(
        public_exponent=65537,
        key_size=3072
    )
    return private_key, private_key.public_key()
```

```
def create_signature(private_key, data: bytes) -> bytes:
    """Create an RSA-PSS digital signature using SHA-256."""
    return private_key.sign(
        data,
        padding.PSS(
            mgf=padding.MGF1(hashes.SHA256()),
            salt_length=padding.PSS.MAX_LENGTH
        ),
        hashes.SHA256()
    )
```

```
def check_signature(public_key, data: bytes, signature: bytes) -> bool:
    """Verify the RSA-PSS digital signature."""
    try:
        public_key.verify(
            signature,
            data,
            padding.PSS(
                mgf=padding.MGF1(hashes.SHA256()),
                salt_length=padding.PSS.MAX_LENGTH
            )
        )
    except InvalidSignature:
        return False
    return True
```

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```
    ),  
    hashes.SHA256()  
)  
    return True  
except InvalidSignature:  
    return False
```

```
private_key, public_key = create_rsa_keys()
```

```
message = b"Payment of 2500 INR has been authorized."
```

```
signature = create_signature(private_key, message)
```

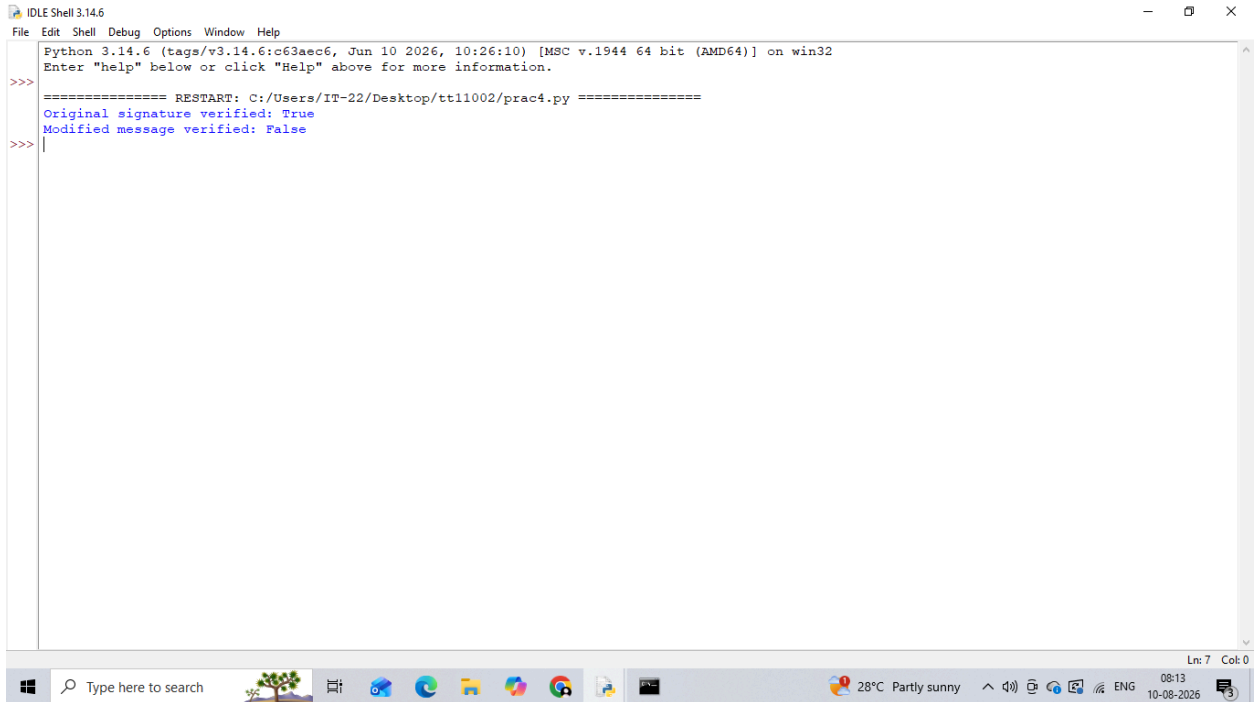
```
print(  
    "Original signature verified:",  
    check_signature(public_key, message, signature)  
)
```

```
changed_message = b"Payment of 7500 INR has been authorized."
```

```
print(  
    "Modified message verified:",  
    check_signature(public_key, changed_message, signature)  
)
```

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OUTPUT :



```
IDLE Shell 3.14.6
File Edit Shell Debug Options Window Help
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/IT-22/Desktop/tt11002/prac4.py =====
Original signature verified: True
Modified message verified: False
>>>
```

GUI CODE :

```
import tkinter as tk
from tkinter import messagebox
from cryptography.hazmat.primitives import hashes
from cryptography.hazmat.primitives.asymmetric import padding, rsa
from cryptography.exceptions import InvalidSignature
```

```
# ----- RSA FUNCTIONS -----
```

```
def create_rsa_keys():
    """Generate RSA private and public keys."""
    private_key = rsa.generate_private_key(
        public_exponent=65537,
        key_size=3072
    )
    return private_key, private_key.public_key()
```

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```
def create_signature(private_key, data):
    """Create RSA-PSS digital signature."""
    return private_key.sign(
        data,
        padding.PSS(
            mgf=padding.MGF1(hashes.SHA256()),
            salt_length=padding.PSS.MAX_LENGTH
        ),
        hashes.SHA256()
    )

def check_signature(public_key, data, signature):
    """Verify the digital signature."""
    try:
        public_key.verify(
            signature,
            data,
            padding.PSS(
                mgf=padding.MGF1(hashes.SHA256()),
                salt_length=padding.PSS.MAX_LENGTH
            ),
            hashes.SHA256()
        )
        return True
    except InvalidSignature:
        return False
```

----- GUI VARIABLES -----

```
private_key = None
public_key = None
signature = None
```

----- GUI FUNCTIONS -----

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```
def generate_keys():
    global private_key, public_key

    private_key, public_key = create_rsa_keys()

    status_label.config(
        text="✓ RSA Keys Generated Successfully",
        fg="#ff69b4"
    )

def sign_message():
    global signature

    if private_key is None:
        messagebox.showwarning(
            "Warning",
            "Please generate RSA keys first."
        )
        return

    message = message_entry.get("1.0", tk.END).strip()

    if not message:
        messagebox.showwarning(
            "Warning",
            "Please enter a message."
        )
        return

    signature = create_signature(
        private_key,
        message.encode()
    )

    signature_box.delete("1.0", tk.END)
```

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```
signature_box.insert(  
    tk.END,  
    signature.hex()  
)
```

```
status_label.config(  
    text="✓ Message Signed Successfully",  
    fg="#ff69b4"  
)
```

```
def verify_message():  
    if public_key is None:  
        messagebox.showwarning(  
            "Warning",  
            "Please generate RSA keys first."  
        )  
    return
```

```
if signature is None:  
    messagebox.showwarning(  
        "Warning",  
        "Please sign a message first."  
    )  
    return
```

```
message = verify_entry.get("1.0", tk.END).strip()
```

```
if not message:  
    messagebox.showwarning(  
        "Warning",  
        "Please enter a message to verify."  
    )  
    return
```

```
valid = check_signature(  
    public_key,
```

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```
message.encode(),
signature
)

if valid:
    result_label.config(
        text="✓ SIGNATURE VALID\n"
        "Message is authentic and unchanged.",
        fg="#00ff88"
    )
else:
    result_label.config(
        text="✗ SIGNATURE INVALID\n"
        "Message has been modified or is not authentic.",
        fg="#ff3366"
    )
```

```
def clear_all():
    message_entry.delete("1.0", tk.END)
    verify_entry.delete("1.0", tk.END)
    signature_box.delete("1.0", tk.END)
```

```
result_label.config(text="")
```

```
status_label.config(
    text="Ready",
    fg="#ff69b4"
)
```

```
# ----- MAIN WINDOW -----
```

```
root = tk.Tk()
```

```
root.title("Digital Signature - PUSHPAL PATHAK")
root.geometry("850x750")
```

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```
root.configure(bg="#0d0d0d")
root.resizable(False, False)
```

```
# ----- TITLE -----
```

```
title_label = tk.Label(
    root,
    text="DIGITAL SIGNATURE",
    font=("Arial", 26, "bold"),
    bg="#0d0d0d",
    fg="#ff69b4"
)
title_label.pack(pady=(20, 15))
```

```
# ----- STATUS -----
```

```
status_label = tk.Label(
    root,
    text="Ready",
    font=("Arial", 11, "bold"),
    bg="#0d0d0d",
    fg="#ff69b4"
)
status_label.pack(pady=5)
```

```
# ----- KEY SECTION -----
```

```
key_frame = tk.Frame(
    root,
    bg="#1a1a1a",
    highlightbackground="#ff69b4",
    highlightthickness=1
)
key_frame.pack(padx=25, fill="x")
```

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```
tk.Label(  
    key_frame,  
    text="RSA KEY GENERATION",  
    font=("Arial", 12, "bold"),  
    bg="#1a1a1a",  
    fg="#ff69b4"  
).pack(pady=10)
```

```
generate_button = tk.Button(  
    key_frame,  
    text="GENERATE RSA KEYS",  
    command=generate_keys,  
    font=("Arial", 10, "bold"),  
    bg="#ff1493",  
    fg="white",  
    activebackground="#ff69b4",  
    activeforeground="black",  
    relief="flat",  
    padx=20,  
    pady=8,  
    cursor="hand2"  
)  
generate_button.pack(pady=(0, 12))
```

----- MESSAGE SECTION -----

```
tk.Label(  
    root,  
    text="ENTER MESSAGE",  
    font=("Arial", 12, "bold"),  
    bg="#0d0d0d",  
    fg="#ff69b4"  
).pack(pady=(15, 5))
```

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```
message_entry = tk.Text(  
    root,  
    height=3,  
    width=90,  
    bg="#1a1a1a",  
    fg="white",  
    insertbackground="#ff69b4",  
    font=("Arial", 10),  
    relief="flat",  
    highlightbackground="#ff69b4",  
    highlightthickness=1  
)  
message_entry.pack(padx=25)
```

----- SIGN BUTTON -----

```
sign_button = tk.Button(  
    root,  
    text="◆ SIGN MESSAGE",  
    command=sign_message,  
    font=("Arial", 10, "bold"),  
    bg="#ff1493",  
    fg="white",  
    activebackground="#ff69b4",  
    activeforeground="black",  
    relief="flat",  
    padx=25,  
    pady=8,  
    cursor="hand2"  
)  
sign_button.pack(pady=12)
```

----- SIGNATURE -----

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```
tk.Label(  
    root,  
    text="DIGITAL SIGNATURE",  
    font=("Arial", 12, "bold"),  
    bg="#0d0d0d",  
    fg="#ff69b4"  
).pack()
```

```
signature_box = tk.Text(  
    root,  
    height=4,  
    width=90,  
    bg="#1a1a1a",  
    fg="#00ff88",  
    font=("Consolas", 8),  
    relief="flat",  
    highlightbackground="#ff69b4",  
    highlightthickness=1  
)  
signature_box.pack(padx=25, pady=5)
```

----- VERIFICATION -----

```
tk.Label(  
    root,  
    text="MESSAGE TO VERIFY",  
    font=("Arial", 12, "bold"),  
    bg="#0d0d0d",  
    fg="#ff69b4"  
).pack(pady=(10, 5))
```

```
verify_entry = tk.Text(  
    root,
```

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```
height=3,  
width=90,  
bg="#1a1a1a",  
fg="white",  
insertbackground="#ff69b4",  
font=("Arial", 10),  
relief="flat",  
highlightbackground="#ff69b4",  
highlightthickness=1  
)  
verify_entry.pack(padx=25)
```

```
verify_button = tk.Button(  
    root,  
    text="✓ VERIFY SIGNATURE",  
    command=verify_message,  
    font=("Arial", 10, "bold"),  
    bg="#c71585",  
    fg="white",  
    activebackground="#ff69b4",  
    activeforeground="black",  
    relief="flat",  
    padx=25,  
    pady=8,  
    cursor="hand2"  
)  
verify_button.pack(pady=10)
```

----- RESULT -----

```
result_label = tk.Label(  
    root,  
    text="",  
    font=("Arial", 11, "bold"),  
    bg="#0d0d0d",
```

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```
    fg="#ff69b4",  
    justify="center"  
)  
result_label.pack(pady=5)
```

----- CLEAR BUTTON -----

```
clear_button = tk.Button(  
    root,  
    text="CLEAR",  
    command=clear_all,  
    font=("Arial", 9, "bold"),  
    bg="#222222",  
    fg="#ff69b4",  
    activebackground="#333333",  
    activeforeground="white",  
    relief="flat",  
    padx=20,  
    pady=6,  
    cursor="hand2"  
)  
clear_button.pack(pady=5)
```

----- FOOTER -----

```
footer = tk.Label(  
    root,  
    text="PUSHPAL PATHAK",  
    font=("Arial", 9),  
    bg="#0d0d0d",  
    fg="#777777"  
)  
footer.pack(side="bottom", pady=10)
```

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----- START GUI -----

root.mainloop()

